



UNIVERSITY OF RUHUNA
BACHELOR OF INFORMATION AND COMMUNICATION TECHNOLOGY
ICT4163 - Digital Image Processing
Practical 05 – Histogram processing



1. Select a suitable grayscale image from the internet and perform the following operations.
 - a) Create a variable named “Image” and read the image to Google Colab environment using open CV.
 - b) Calculate the histogram of the image.
 - c) Display the original image using a plot.
 - d) Display the histogram of the original image.
 - e) Apply histogram equalization to the original image.
 - f) Display the Original image, histogram equalized image.
 - g) Calculate the histogram of the histogram equalized image.
 - h) Display the histogram of the histogram equalized image.
2. Select a suitable grayscale image from the internet and perform the following operations.
 - a) Create a variable named “ClaheImage” and read the image to Google Colab environment using open CV.
 - b) Create a CLAHE object with the following configurations.
 - a. clipLimit=2.0
 - b. tileGridSize=(8,8)
 - c) Apply Contrast Limited Adaptive Histogram Equalization to the original image.
 - d) Display the enhanced image after applying CLAHE.
3. Select a low contrast color image from the internet and perform the following operations.
 - a) Create a variable named “Image_LC” and Read the image to Google Colab environment using open CV.
 - b) Separate the color channels of the image.
 - c) Calculate and display the histogram of each color channel of the image.
 - d) Perform the histogram equalization on each channel of the image.
 - e) Calculate and display the histogram of each equalized color channel of the image.
 - f) Merge the color channels and create new image and display.